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COVID-19 Survival Prediction

Introduction

The Covid-19 virus began to spread in 2019 with a first confirmed case in the United States in early 2020. Survivability was a huge question originally with many reported cases of death.

In order to provide better care, it would help if we knew what symptoms are the leading cause of death in covid patients.

What symptoms cause the most death in covid patients?

The Data

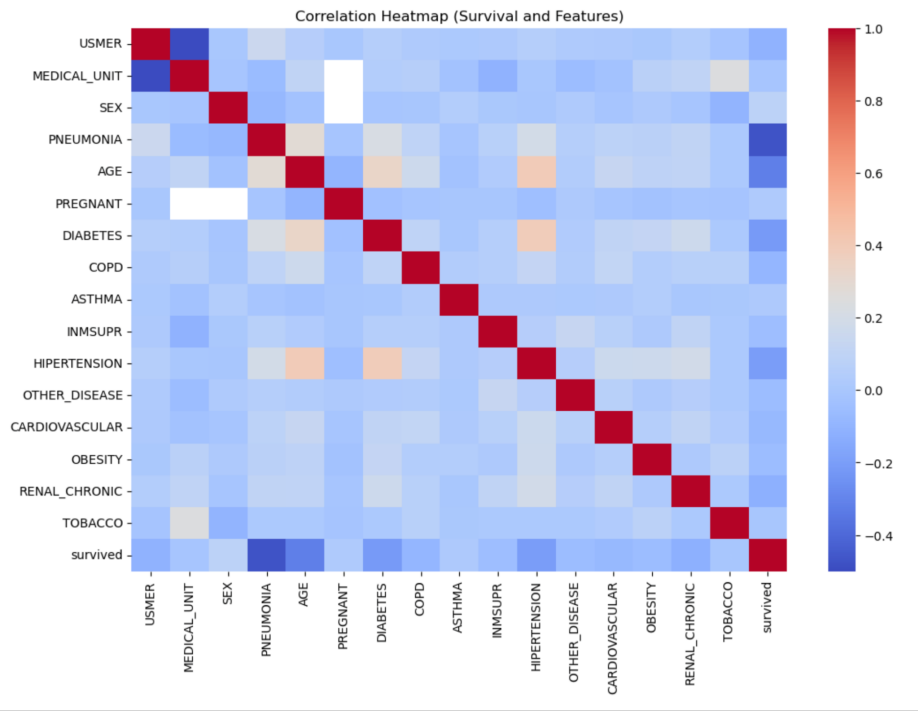
- Covid-19 data set found on here on Kaggle.
- 1,048,576 observations
- Provided by the Mexican government.
- Mostly boolean features
- 21 features

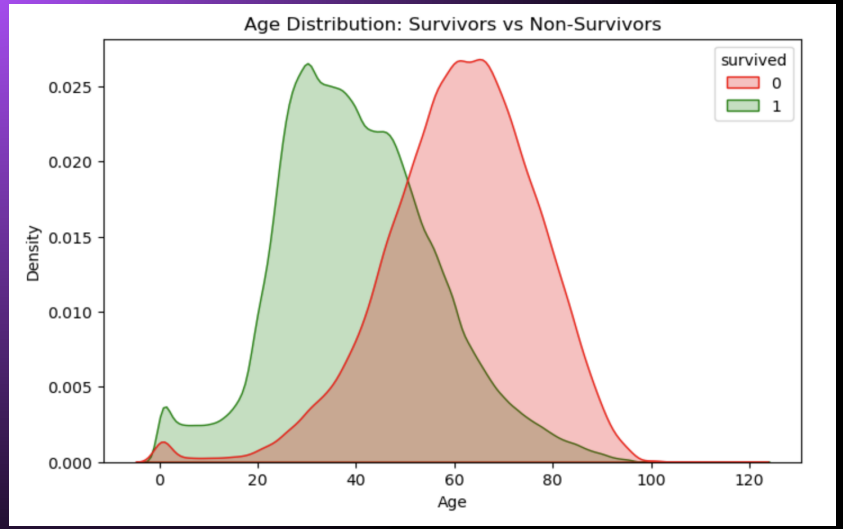
Features

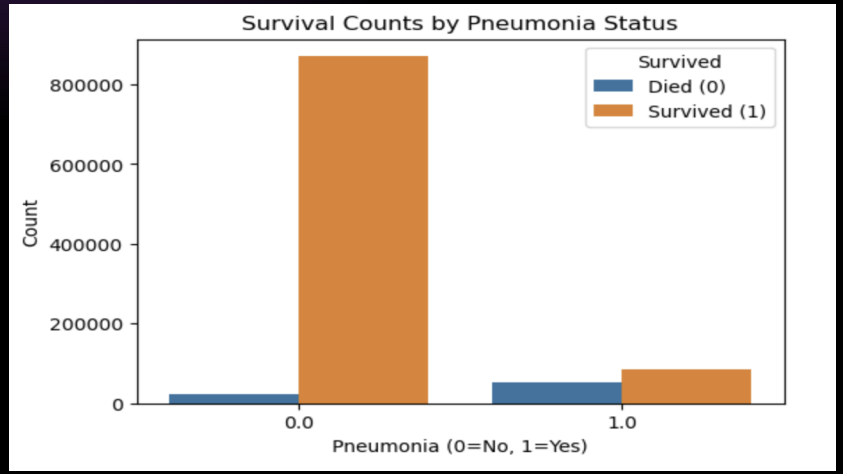
(some)

- Age
- Sex
- Tobacco
- Pneumonia
- Renal Chronic
- Obesity
- Immunosuppressed
- Asthma
- COPD
- Diabetes
- Hypertension
- Pregnancy
- Date Died

Exploratory Data Analysis





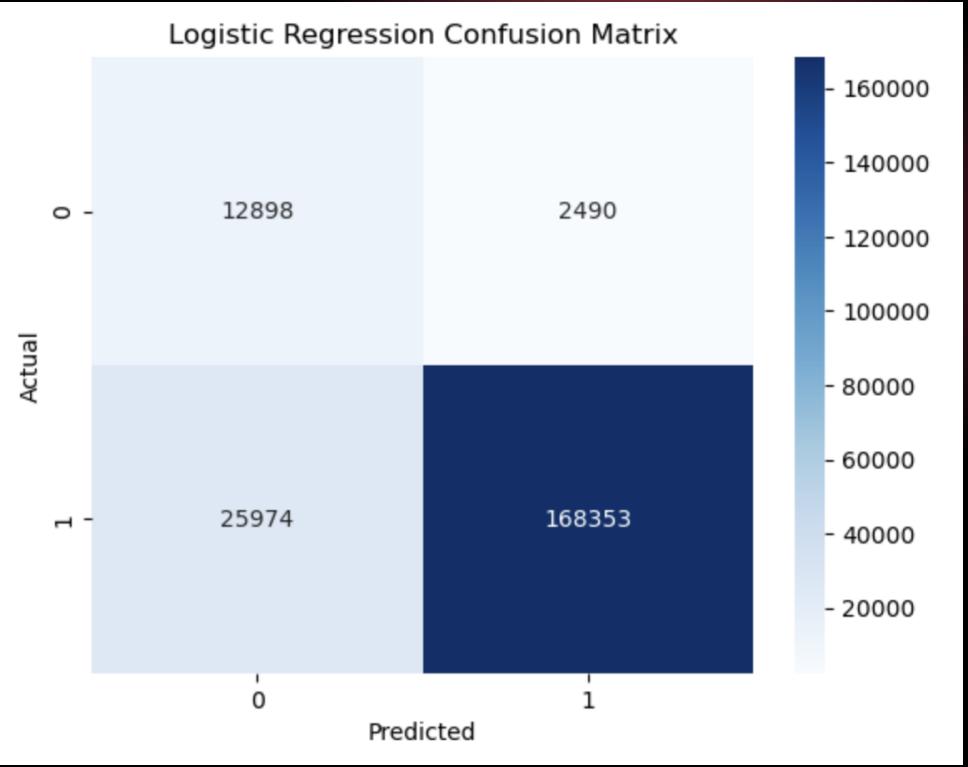


Final Model

The final model chosen was Logistic Regression which distinguished survivors vs. death 92% of the time, compared to Random Forest Classification model only making the correct distinction 87% of the time.

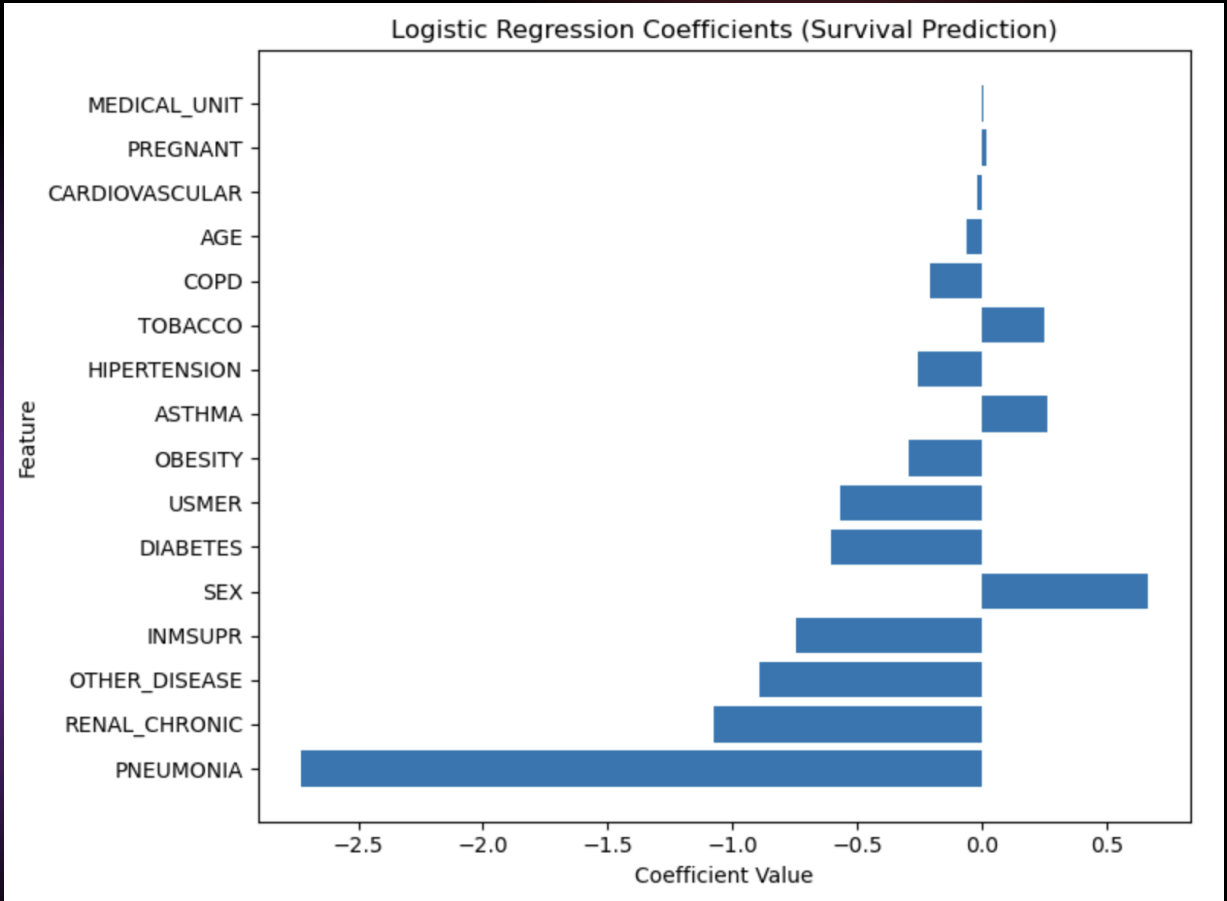
	precision	recall	f1-score	support
0	0.33	0.84	0.48	15388
1	0.99	0.87	0.92	194327
accuracy			0.86	209715
macro avg	0.66	0.85	0.70	209715
weighted avg	0.94	0.86	0.89	209715

ROC AUC: 0.9214003608842594



Results

We can see the leading predictor of death was pneumonia, as well as chronic kidney disease and being immunosuppressed.



Conclusions & Future Work

In conclusion, we can see that the feature that had the most impact on death of a patient was Pneumonia.

Other factors include being immunosuppressed, kidney disease, as well as other diseases (undisclosed).

Future work may include:

- Testing other models
- More specific research question

THANK YOU

Any questions?